

Question	Working	Answer	Mark	Notes		
15	(a)	Shown	M1	for method to establish at least one root between $x = 0$ and $x = 1$, eg $f(0) = -5$ and $f(1) = 3$		
			C1	for correct values and a deduction about the roots eg as there is a sign change there must be at least one root between $x = 0$ and $x = 1$ (as f is continuous)		
		Shown	C1	for a correct first step in rearrangement, eg $x(x^2 + 7) - 5 = 0$ or $x^3 + 7x = 5$		
			C1	for clear and correct steps showing complete rearrangement		
		(c)	$x_1 = 0.625$ $x_2 = 0.6765327696$ $x_3 = 0.6704483001$	0.6704(483001)	M1	for substitution of 1 into the formula (to get 0.625)
					M1	for substitution of “ $x_1 = 0.625$ ” and “ $x_2 = 0.6765327696$ ” to give x_2 and x_3
					A1	0.6704(483001)
		(d)	Comment	M1	substitutes answer to (c) into expression (to get $-0.00549\dots$)	
C1	appropriate comment, eg accurate as answer is close to 0					